

AI Project 4 Answer Doc

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Question 4:

- a) The probability that the first player wins assuming both players play optimally from the start is calculated by our algorithm. When our current score, $i = 0$ and the opponent's score, $j = 0$, and the current turn, $k = 0$, then the probability of the first player winning is 0.5305927250129694. We define optimality here by assuming that all of a player's rolls will be in favor of them winning starting from the first turn. Essentially, when the game has just started and both the player and their opponent's score is 0, and player 1 gets the best rolls, the chance to win is 0.5305927250129694.
- b) On the first turn, the lowest turn total where the player should hold rather than roll is 21. This is determined by the algorithm which calculates the probabilities of winning and chooses the max.
- c) If player 1 scores no points on their first turn the probability that the second player wins is the same as with 4a, 0.5305927250129694. This is because the same conditions apply where player 1 now has no points and player 2 plays optimally from their turn 1. For the same reason, the hold threshold for player 2 matches 4b, as it is also 21.